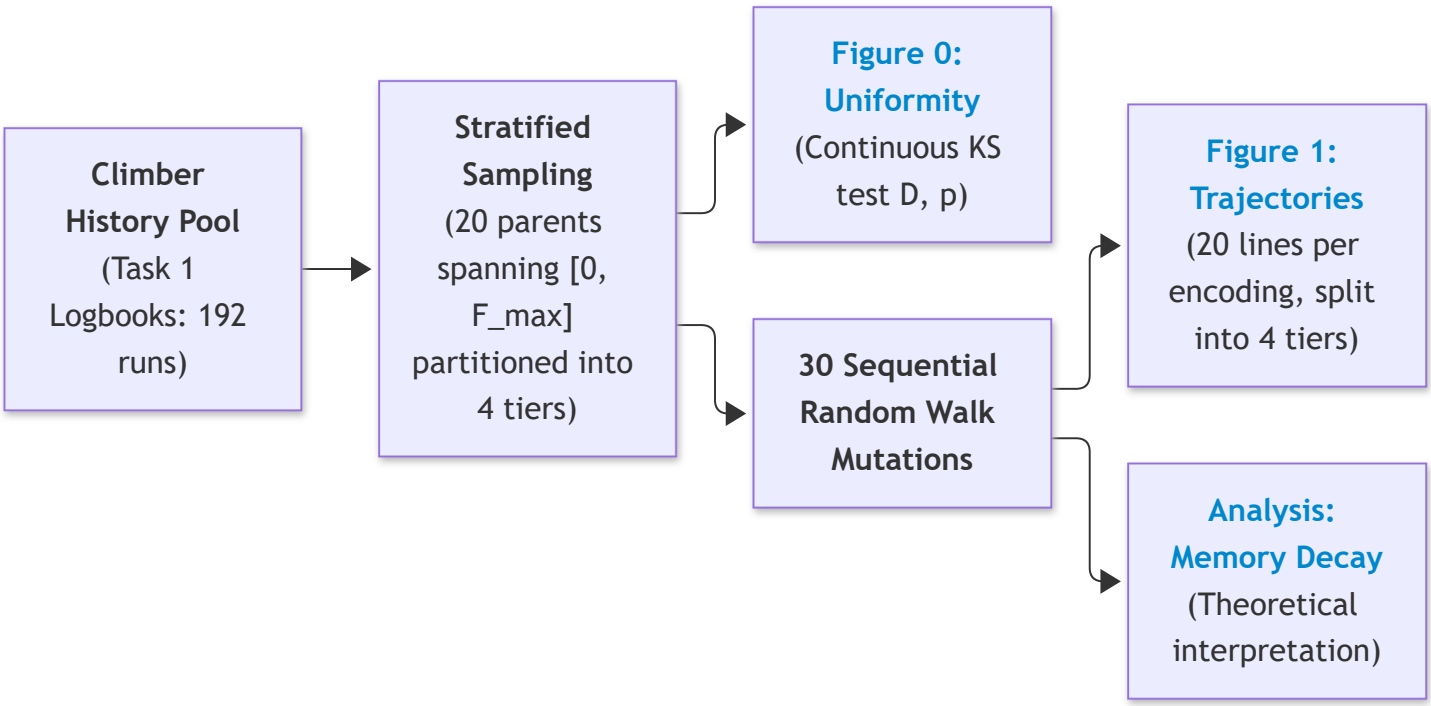
Assignment 6: Fitness Landscape Exploration

Task 3: Random Walk Fitness Landscape Exploration

Objective

The objective of this task is to investigate the correlation length, mutational fragility, and genetic memory of the **fitness landscape for net rectilinear velocity** across different fitness levels for four genetic encodings (f_0, f_1, f_4, f_H).

Experiment Design & Pipeline



For each representation, $N = 20$ parent solutions are selected across the full discovered fitness spectrum $[0, F_{\max}]$, as it's done in Task 1: fitness spectrum $[0, F_{\max}]$ is binned into $n_{\text{bins}} = 4$ equal-width bins with $k = 5$ samples per each bin, which naturally correspond to initial quality tiers. Starting from each solution, a sequential chained random walk of 30 mutations is executed, rejecting infeasible mutants, to examine how random mutational drift unfolds at different fitness levels and measure the decay of evolved genetic information ("solution as memory").

Results

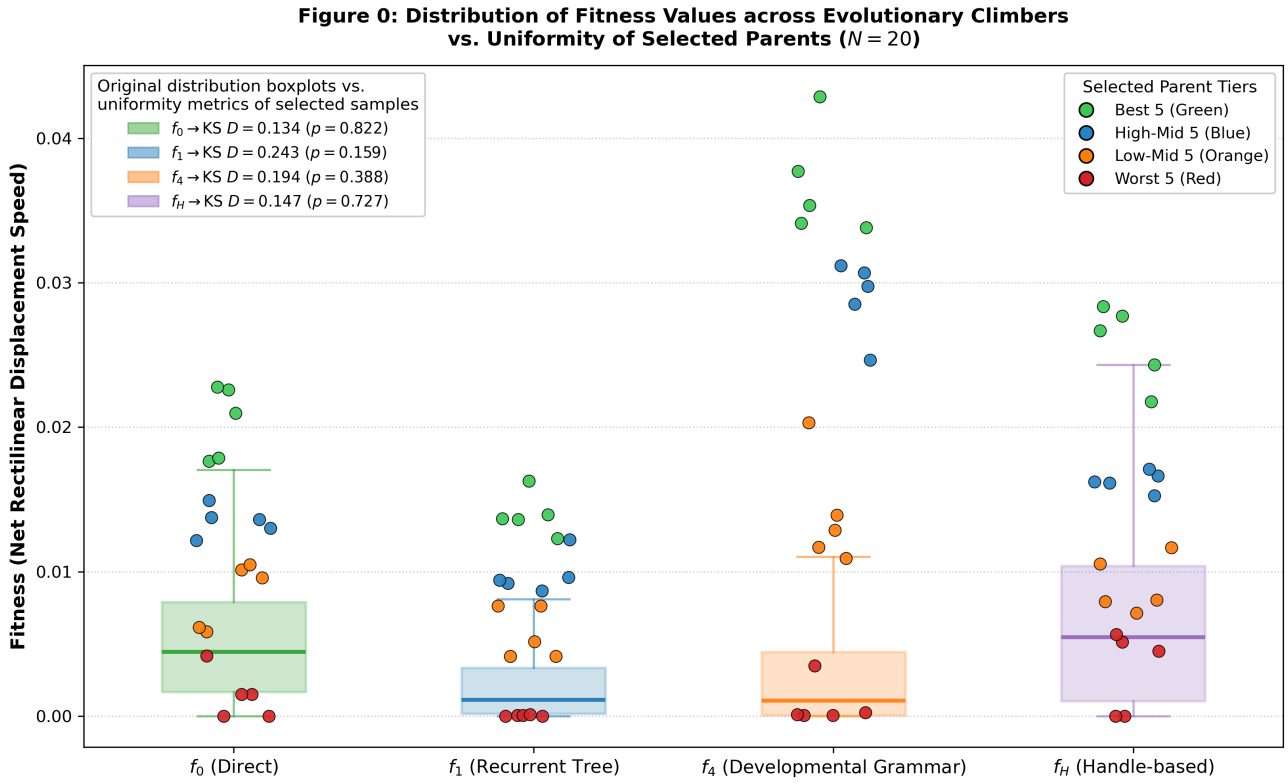


Figure 0: *Distribution of fitness values across unique evolutionary climbers (semi-transparent boxplots correspond to Task 1: Figure 0 distributions) vs. the 20 stratified parent solutions (overlaid points colored by quality tier (5 per each)). Left legend reports the continuous 1-sample Kolmogorov-Smirnov test against theoretical uniform distribution* $\mathcal{U}[0, F_{\max}^{\text{enc}}]$ *for the 20 selected parents for each encoding* enc *, confirming uniformity of the selected parent fitness values distribution.*

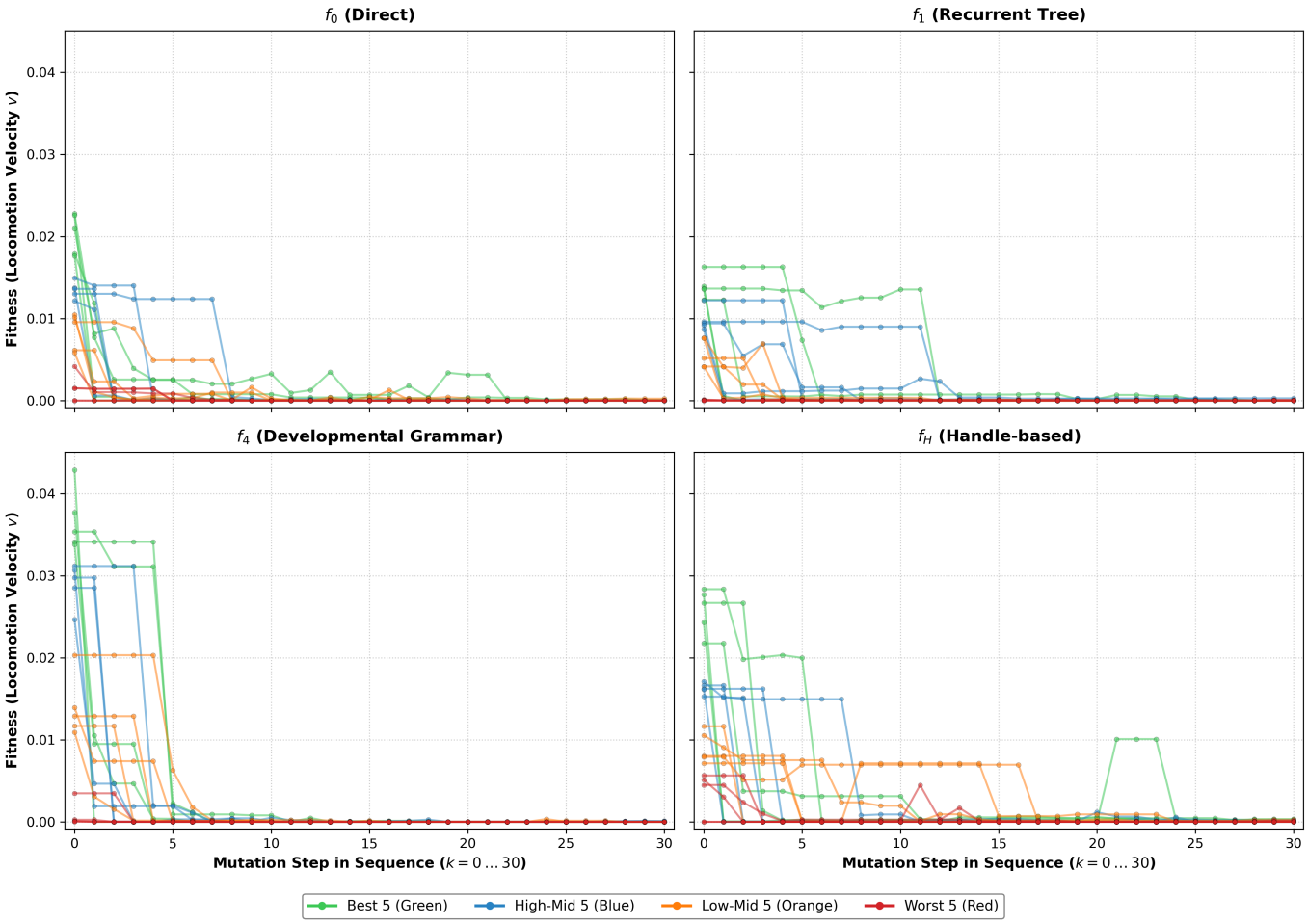


***Figure 1**: Sequential Random Walk Fitness Trajectories across 4 encodings. Chained mutations are initiated from 20 parent solutions per representation, and colored by their fitness tier (worst to best)*:

- f_0 exhibits rapid additive collapse as random perturbations break mechanical joints or neural connections.
- f_1 displays subtree modularity, where mutations either maintain neutrality (since most of them - 67.1% - apply to the neural net) or cause abrupt discrete drops when a branching node is severed; this can be compared to the stagnation effect, observed in Task 1.
- f_4 demonstrates rule-neutral plateaus followed by catastrophic morphospace collapse once a core developmental rule is altered.
- f_H exhibits high initial sensitivity of the *top-tier* solutions due to coordinate handle mismatches in Euclidean binding space; *mid-tier* solutions show **robustness** to mutations (probably due to the threshold effect, specific to this grammar) - similar effect exhibited in Task 1 - moreover, this also explains emergence of sudden peaks (as on the 21st iteration).

"Solution as Memory" & Landscape Topology

1. **Decoherence of Genetic Memory**: Evolved high-fitness genotypes represent specialized "memory" storing phenotypic coordination discovered under selective pressure. In the absence of selection, unconstrained random walks produce exponential memory decay: fitness drops below 50% within $\approx 2..5$ mutational steps, collapsing to near zero ($< 10^{-4}$) by step 30 across all encodings.
2. **Low-Fitness Basin Entrapment**: Trajectories starting in low-fitness regions (red lines, Tier 4 on Figure 1) remain confined to near-zero fitness throughout the entire 30-step walk. Without directional selection, purely stochastic drift cannot spontaneously assemble the synchronized mechanical and neural coordination required for locomotion.
3. **Landscape Ruggedness**: High-fitness peaks are surrounded by steep fitness cliffs (e.g. top-tier solutions in f_1 on Figure 1). While developmental encodings (f_1, f_4) occasionally sustain neutral plateaus for a few steps, any structural perturbation to critical developmental rules or modular branches results in catastrophic collapse to the non-motile baseline (similar effect was observed in crossover in Task 2: Figure 3).

In general, the trajectories directly reflect the operator-level properties discovered during 1-step neighborhood probing in Task 1: across all encodings, mutation operators exhibit a **severe destructive bias**, where worse mutations (54%–62%) drastically outnumber beneficial ones (1.5%–3.7%), while 35%–58% remain functionally neutral due to silent neural redundancy (Task 1: Neural Redundancy). In chained random walks without selection, this asymmetry makes decay inevitable: continuous parametric shifts (f_0) induce gradual additive drift, discrete tree-node or grammar rewrite mutations (f_1, f_4) act as macro-mutations causing step-function drops, and Euclidean distance thresholds (f_H) provide initial phenotypic buffering against handle perturbations before experiencing threshold-crossing breakdown.